

# AWS Elastic File System (EFS)

## Master Study Notes & Practical Demo Guide

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# 1. What is Amazon EFS?

Amazon Elastic File System (EFS) is a fully managed shared file storage service for Linux workloads in AWS.

It allows multiple EC2 instances to simultaneously access the same files and directories.

EFS uses the Network File System (NFS) protocol.

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## Simple Explanation

Think of EFS like:

- A shared office network drive
  - Accessible from many servers simultaneously
  - Automatically scalable
  - Highly available
- 

## Key Characteristics

| Feature         | Description  |
|-----------------|--------------|
| Storage Type    | File Storage |
| Access Protocol | NFSv4        |
| Shared Access   | Yes          |
| Multi-AZ        | Yes          |
| Managed by AWS  | Yes          |
| Linux Support   | Native       |
| Scaling         | Automatic    |

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# 2. Why EFS Exists

EBS is usually attached to one EC2 instance.

Many applications need:

- Shared storage
- Centralized files
- Multi-server access
- Shared uploads

- Shared application data

EFS solves this problem.

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## 3. Storage Types — EBS vs EFS vs S3

| Feature               | EBS          | EFS          | S3           |
|-----------------------|--------------|--------------|--------------|
| Storage Type          | Block        | File         | Object       |
| Shared Access         | Limited      | Yes          | Via APIs     |
| Mountable             | Yes          | Yes          | No           |
| Multi-Instance Access | Limited      | Yes          | Indirect     |
| Protocol              | Block Device | NFS          | HTTP/API     |
| Best Use              | OS, DB       | Shared Files | Static Files |
| Persistence           | Yes          | Yes          | Yes          |
| Multi-AZ              | No           | Yes          | Yes          |

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## 4. EFS Architecture

### High-Level Flow

EC2 Instance ↓ NFS Client ↓ Mount Target ↓ EFS File System

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### Important Concepts

#### Regional Service

EFS is regional.

#### Multi-AZ

EFS automatically stores data redundantly across multiple Availability Zones.

#### Mount Targets

EFS creates network endpoints inside subnets.

One mount target is created per Availability Zone.

## Network File System

EFS uses NFS protocol over TCP port 2049.

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## 5. Core Components of EFS

### File System

Main EFS storage resource.

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### Mount Targets

Network interfaces inside subnets.

Purpose:

- EC2 connects to EFS through mount targets
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### Security Groups

Control network access to EFS.

Required port:

TCP 2049

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### Access Points

Application-specific entry points into EFS.

Used for:

- Containers
  - Kubernetes
  - Application isolation
-

## 6. How EFS Works Internally

### NFS Protocol

EFS uses:

NFSv4

This allows Linux systems to mount EFS like a local folder.

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### Client-Server Model

EC2 acts as:

- NFS Client

EFS acts as:

- NFS Server
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### Serverless Storage

AWS manages:

- Scaling
  - Replication
  - Redundancy
  - Hardware
  - Disk failures
  - Maintenance
- 

## 7. EFS Performance Modes

### General Purpose

Default mode.

Best for:

- Web applications
- CMS
- Shared files

- General workloads

Low latency.

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## Max I/O

Designed for:

- Massive parallel workloads
- Big data
- Analytics

Higher throughput. Higher latency.

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## 8. Throughput Modes

### Bursting Throughput

Default mode.

Performance scales with storage size.

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### Provisioned Throughput

Manually set throughput.

Used when:

- Small storage
  - High throughput required
- 

### Elastic Throughput

AWS automatically scales throughput.

Recommended modern option.

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## 9. Storage Classes

### Standard

Frequently accessed files.

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### Infrequent Access (IA)

Lower-cost storage.

Automatically moves old files.

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### Archive

Very low-cost long-term storage.

Higher retrieval latency.

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## 10. Security in EFS

### Security Groups

Allow:

TCP 2049

Between:

- EC2
  - EFS Mount Target
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### Encryption at Rest

Uses:

- AWS KMS
  - AES-256
-

## Encryption in Transit

TLS encryption during mount.

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## POSIX Permissions

Linux file permissions still apply:

`chmod`  
`chown`

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## 11. EFS Pricing

Pricing based on:

- Storage used
  - Throughput mode
  - IA storage
  - Data transfer
- 

## Cost Optimization

Use lifecycle policies:

Standard → IA → Archive

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## 12. EFS Use Cases

### Shared Web Content

Multiple web servers sharing files.

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### WordPress Shared Uploads

All instances share:

`/wp-content/uploads`

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## Containers

Used with:

- ECS
- EKS
- Kubernetes
- Docker

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## Shared Home Directories

Linux user directories.

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## Machine Learning

Shared datasets across nodes.

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# 13. EFS Limitations

## Higher Latency than EBS

Because it is network-based.

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## Primarily Linux-Focused

Native Linux integration.

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## Requires Network Connectivity

Cannot work without network access.

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## Not Ideal for Databases

Databases generally prefer EBS.

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## 14. Demo 1 — Create an EFS File System

### Goal

Understand EFS creation and architecture.

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### Console Path

AWS Console → EFS → Create File System

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### Step 1 — Open EFS Console

Go to:

AWS Console → EFS

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### Step 2 — Create File System

Click:

Create file system

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### Step 3 — Configure

Choose:

- VPC
  - Availability Zones
  - Subnets
  - Security Groups
- 

### Step 4 — Enable Encryption

Enable:

- Encryption at rest

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## Step 5 — Review Mount Targets

Observe:

- One mount target per AZ

Very important concept.

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## Step 6 — Create File System

Click:

Create

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# 15. Demo 2 — Mount EFS to an EC2 Instance

## Goal

Connect Linux EC2 instance to EFS.

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## Architecture

EC2 → NFS → EFS

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## Step 1 — Launch EC2 Instance

Launch:

- Amazon Linux 2
  - Same VPC as EFS
- 

## Step 2 — Configure Security Group

Allow:

| Port | Protocol | Purpose |
|------|----------|---------|
| 22   | TCP      | SSH     |

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| Port | Protocol | Purpose |
|------|----------|---------|
| 2049 | TCP      | NFS     |

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## Step 3 — Connect via SSH

```
ssh -i key.pem ec2-user@PUBLIC-IP
```

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## Step 4 — Install EFS Utilities

Amazon Linux:

```
sudo yum install -y amazon-efs-utils
```

Ubuntu:

```
sudo apt update  
sudo apt install -y amazon-efs-utils
```

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## Step 5 — Create Mount Directory

```
sudo mkdir /efs
```

---

## Step 6 — Mount EFS

```
sudo mount -t efs fs-xxxxxxx:/ /efs
```

---

## Step 7 — Verify Mount

```
df -h
```

You should see EFS mounted.

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# 16. Demo 3 — Shared Storage Across Multiple EC2 Instances

## Goal

Demonstrate shared storage.

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## Architecture

EC2-1 EC2-2 ↓ Shared EFS

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### Step 1 — Launch Two EC2 Instances

Create:

- EC2-1
  - EC2-2
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### Step 2 — Mount Same EFS on Both

Mount EFS using same mount command.

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### Step 3 — Create File on EC2-1

```
echo "Hello from EC2-1" > /efs/test.txt
```

---

### Step 4 — Verify on EC2-2

```
cat /efs/test.txt
```

File appears immediately.

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## Important Learning

This is why EFS is used for:

- Shared application storage
  - CMS uploads
  - Shared content
  - Containers
-

## 17. Demo 4 — Persistent Shared Web Storage

### Goal

Simulate enterprise architecture.

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### Architecture

ALB ↓ EC2-1 EC2-2 ↓ Shared EFS

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### Step 1 — Install Apache

```
sudo yum install -y httpd
sudo systemctl enable httpd
sudo systemctl start httpd
```

---

### Step 2 — Mount EFS to Web Directory

```
sudo mount -t efs fs-xxxxxxx:/ /var/www/html/uploads
```

---

### Step 3 — Create File on EC2-1

```
echo "Shared Storage" > /var/www/html/uploads/index.html
```

---

### Step 4 — Access from EC2-2

Same file visible.

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## 18. Demo 5 — Auto-Mount EFS on Reboot

### Goal

Persist mount after reboot.

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## Step 1 — Edit fstab

```
sudo nano /etc/fstab
```

---

## Step 2 — Add Entry

```
fs-xxxxxxx:/ /efs efs defaults,_netdev 0 0
```

---

## Step 3 — Save File

Ctrl+O Enter Ctrl+X

---

## Step 4 — Test Configuration

```
sudo mount -a
```

---

## Step 5 — Reboot

```
sudo reboot
```

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## Step 6 — Verify

```
df -h
```

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# 19. Demo 6 — Security Group Failure Troubleshooting

## Goal

Understand NFS networking.

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## Step 1 — Remove Port 2049

Remove:

TCP 2049

From EFS security group.

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## Step 2 — Attempt Mount

Mount fails.

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## Step 3 — Restore Port 2049

Add:

TCP 2049

Mount works again.

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## Learning Outcome

EFS requires:

NFS over TCP 2049

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# 20. Demo 7 — Encryption in EFS

## Goal

Understand encrypted storage.

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## Step 1 — Create Encrypted EFS

Enable:

Encryption at rest.

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## Step 2 — Mount Normally

No application changes required.

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## Step 3 — Explain

AWS automatically encrypts:

- Stored files
  - Metadata
  - Snapshots
- 

## 21. Demo 8 — Lifecycle Management & Infrequent Access

### Goal

Cost optimization.

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### Step 1 — Open Lifecycle Policies

EFS → Lifecycle Management

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### Step 2 — Enable Transition

Move files:

Standard → IA

After:

30 days

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### Step 3 — Explain

Old files automatically move to cheaper storage.

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## 22. Demo 9 — Multi-AZ High Availability

### Goal

Understand EFS resilience.

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### Step 1 — Open Mount Targets

Observe:

- Mount target in multiple AZs
- 

### Step 2 — Explain

If one AZ fails:

- EFS still available
  - Other mount targets continue serving traffic
- 

## 23. Monitoring EFS with CloudWatch

### Important Metrics

| Metric             | Meaning                      |
|--------------------|------------------------------|
| BurstCreditBalance | Remaining burst credits      |
| PercentIOLimit     | Approaching throughput limit |
| DataReadIOBytes    | Read traffic                 |
| DataWriteIOBytes   | Write traffic                |
| ClientConnections  | Number of clients            |

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### Monitoring Best Practices

Configure alarms for:

- High throughput
- Low burst credits
- High latency

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## 24. Enterprise Architecture Patterns

### High Availability WordPress

CloudFront ↓ ALB ↓ Auto Scaling EC2 ↓ Shared EFS

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### CI/CD Platform

Jenkins master + agents share:

`/var/lib/jenkins`

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### Machine Learning Platform

GPU nodes share datasets using EFS.

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### Shared Container Storage

Multiple ECS tasks mount same EFS.

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## 25. Best Practices

### Use Lifecycle Policies

Reduce storage cost.

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### Use Security Groups Properly

Allow only required access.

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## Enable Encryption

Protect sensitive data.

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## Use Access Points

Application isolation.

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## Use Auto-Mount

Ensure persistence after reboot.

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## Monitor Throughput

Avoid performance bottlenecks.

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## 26. Common Interview Questions

### What protocol does EFS use?

NFSv4.

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### What port does EFS use?

TCP 2049.

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### Why use EFS instead of EBS?

Shared access.

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### Why not use EFS for databases?

Higher network latency.

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## Is EFS multi-AZ?

Yes.

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## Can multiple EC2 instances mount EFS simultaneously?

Yes.

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# 27. Troubleshooting Guide

## Mount Timeout

Possible causes:

- Security group issue
  - Port 2049 blocked
  - Wrong subnet
  - Wrong VPC
- 

## Permission Denied

Check:

```
ls -la  
chmod  
chown
```

---

## DNS Resolution Failure

Verify:

- VPC DNS enabled
  - Correct EFS DNS name
- 

## Slow Performance

Check:

- Throughput mode

- Burst credits
  - CloudWatch metrics
- 

## 28. Quick Reference Cheat Sheet

### Important Concepts

| Term         | Meaning                |
|--------------|------------------------|
| EFS          | Elastic File System    |
| NFS          | Network File System    |
| Mount Target | Network endpoint       |
| Access Point | App-specific EFS entry |
| IA           | Infrequent Access      |
| KMS          | Key Management Service |

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### Important Ports

| Port | Purpose |
|------|---------|
| 22   | SSH     |
| 2049 | NFS     |

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### Console Paths

| Task             | Console Path               |
|------------------|----------------------------|
| Create EFS       | EFS → Create File System   |
| Mount Targets    | EFS → Network              |
| Lifecycle Policy | EFS → Lifecycle Management |
| Monitoring       | CloudWatch → Metrics → EFS |

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## 29. Important Linux Commands

### Install EFS Utilities

Amazon Linux:

```
sudo yum install -y amazon-efs-utils
```

Ubuntu:

```
sudo apt install -y amazon-efs-utils
```

---

## Create Directory

```
sudo mkdir /efs
```

---

## Mount EFS

```
sudo mount -t efs fs-xxxxxxx:/ /efs
```

---

## Verify Mount

```
df -h
```

---

## Permanent Mount

```
sudo nano /etc/fstab
```

Add:

```
fs-xxxxxxx:/ /efs efs defaults,_netdev 0 0
```

---

## Test fstab

```
sudo mount -a
```

---

## Check Mounted Filesystems

```
mount | grep efs
```

---

## Final Summary

Amazon EFS is a fully managed, scalable, highly available shared file storage service for Linux workloads in AWS.

It enables multiple EC2 instances, containers, and applications to access the same shared filesystem simultaneously using NFS.

EFS is commonly used in:

- Shared web hosting
- Containers
- Kubernetes
- CI/CD pipelines
- Machine learning platforms
- Enterprise shared storage architectures